

Definition and first terms

$8\,A(x)=7+8\,x+A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(2)*T(x)$; $T=x+3*T^2+4*T^3+2*T^4$.
 $a(0),\ldots,a(7)=1,2,6,44,394,3948,42364,476120$.
 Kernel used throughout: $\rho(u)=u\,(1-3\,u^1-4\,u^2-2\,u^3)$, so $x=\rho(T(x))$.

Typogeometry

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2\colon 3$, $\Delta_3\colon 4$, $\Delta_4\colon 2$.

Coefficient and contour extraction

$$\mathcal{K}_n=\{(k_1,\ldots,k_3)\in\mathbb{Z}_{\geq 0}^3:\sum_{j=1}^3jk_j=n-1\},\quad K=\sum_{j=1}^3k_j,$$

$$a(n)=\frac{2}{n}\sum_{\boldsymbol{k}\in\mathcal{K}_n}\binom{n+K-1}{K}\binom{K}{k_1,\ldots,k_3}3^{k_1}4^{k_2}2^{k_3}.$$

$$a(n)=\frac{2}{2\pi i n}\oint_{\gamma}\frac{du}{\rho(u)^n},\quad n\geq 1.$$

Recurrence

$$\sum_{r=0}^3P_r(n)a(n+r)=0,\,n\geq 1,\,\text{with}$$

$$\begin{aligned}P_0(n)&=-512\,n^3-768\,n^2-96\,n+80\\P_1(n)&=-1344\,n^3-4032\,n^2-3892\,n-1204\\P_2(n)&=-1176\,n^3-5292\,n^2-7644\,n-3528\\P_3(n)&=89\,n^3+534\,n^2+979\,n+534\end{aligned}$$

Differential equation

$$\begin{aligned}(80\,x^3)\,A(x)+(-1376\,x^4-1204\,x^3)\,A'(x)\\+(-2304\,x^5-4032\,x^4-1764\,x^3)\,A^{(2)}(x)+(-512\,x^6-1344\,x^5-1176\,x^4+89\,x^3)\,A^{(3)}(x)=0.\end{aligned}$$

Matrices, telescoper certificate, and checks

$$U=\begin{pmatrix}\frac{1016}{89}&\frac{144}{89}&\frac{12}{89}&\frac{1}{89}\\ \frac{1824}{89}&\frac{152}{89}&\frac{72}{89}&\frac{6}{89}\\ \frac{1152}{89}&\frac{96}{89}&\frac{8}{89}&\frac{60}{89}\\ 0&0&0&0\end{pmatrix},\quad V=\begin{pmatrix}-1&0&0&0\\ \frac{482}{89}&\frac{55}{89}&\frac{12}{89}&\frac{1}{89}\\ \frac{600}{89}&\frac{50}{89}&\frac{19}{89}&\frac{9}{89}\\ \frac{288}{89}&\frac{24}{89}&\frac{2}{89}&\frac{15}{89}\end{pmatrix},\quad J=\begin{pmatrix}0&1&0&0\\ 0&0&2&0\\ 0&0&0&3\\ 0&0&0&0\end{pmatrix}$$

Certificate.

$$R(n,u)=N(n,u)/\rho(u)^2,\,\deg_u N=9.$$

Telescoping identity: $\sum_rP_r(n)H_{n+r}(u)=\partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.